

DORI

Hardware, Calibration & Final Prototype

Every figure, table and calibration equation — Chapters 2–4 and Appendix

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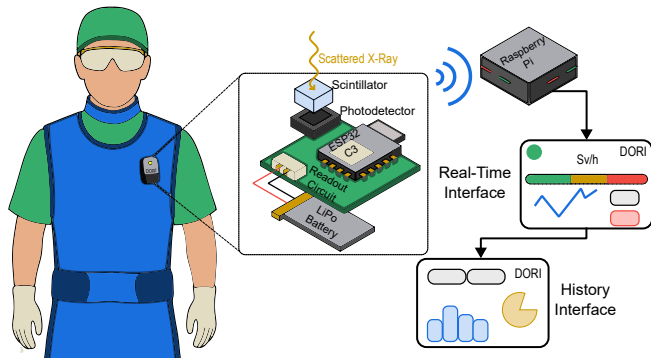
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Chapter 2

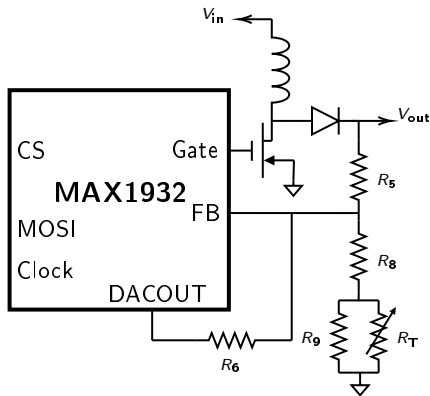
Hardware Development

Conceptual Design of DORI



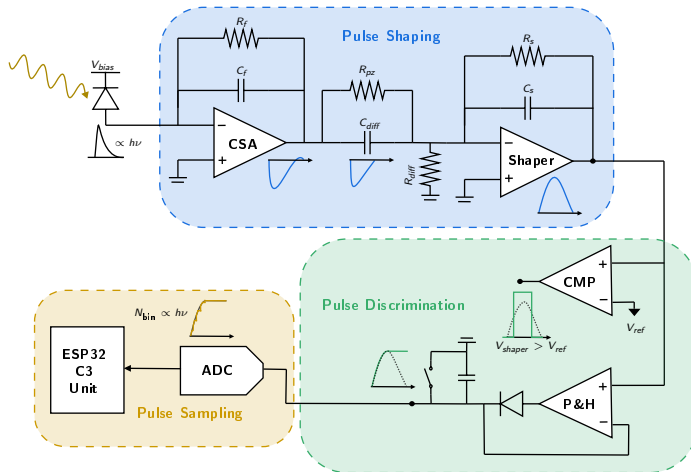
The wearable unit (exploded view) is worn over the protective apron and comprises the scintillator, photodetector, readout circuit, MCU and LiPo battery. Data are sent to a Raspberry Pi, which displays the real-time and history interfaces.

SiPM Bias Supply

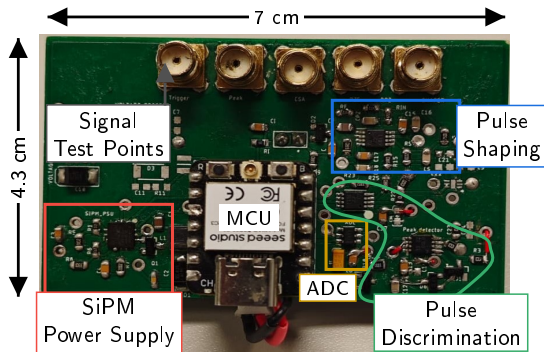


MAX1932 feedback network with the NTC thermistor R_T for temperature compensation of the SiPM bias. Pin labels on the left correspond to SPI connections.

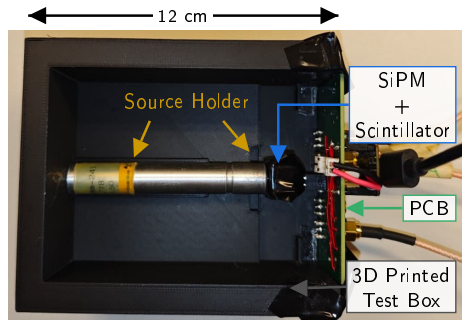
Architecture of the DORI System



DORI-Test Board



(a)



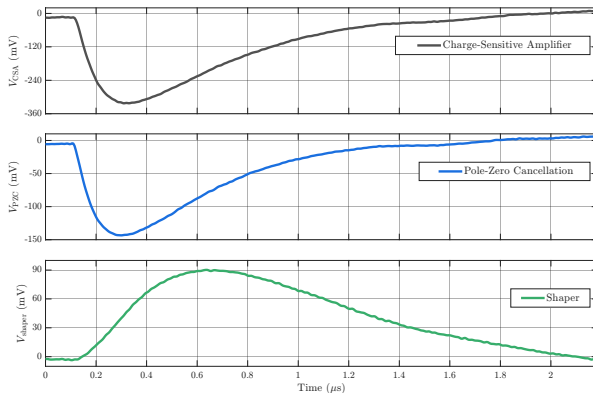
(b)

(a) PCB with the main functional blocks identified, together with the signal test points used for characterisation. (b) Assembled prototype inside the 3D printed test box, showing the sealed source holder aligned with the scintillator and SiPM.

Component Values of the Pulse Shaping Chain

CSA		PZC		Shaper	
Parameter	Value	Parameter	Value	Parameter	Value
R_f	4.7 k Ω	R_{diff}	4.7 k Ω	R_s	10 k Ω
C_f	100 pF	R_{pz}	4.7 k Ω	C_s	47 pF
		C_{pz}	100 pF		
τ	470 ns		470 ns		470 ns

Transient Response of the Pulse Shaping Chain

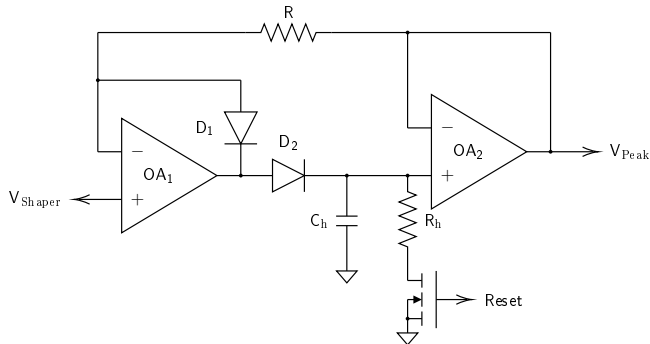


Transient response of the CSA (grey), PZC (blue) and shaper (green) to a single ^{241}Am event.

Timing Metrics of the Pulse Shaping Chain

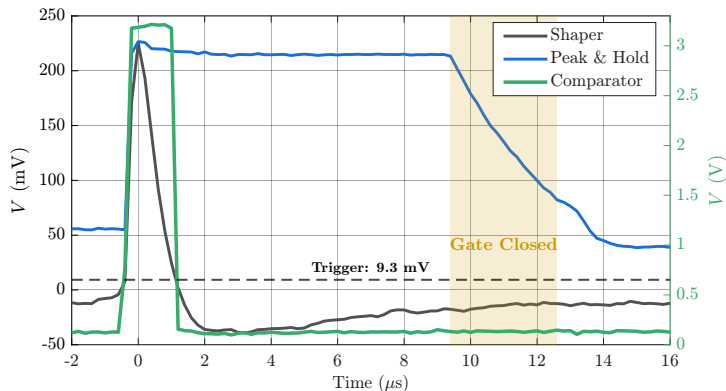
	CSA	PZC	Shaper
Amplitude range (mV)	191–861	87–430	61–407
Peaking time (ns)	147 ± 28	135 ± 25	387 ± 50
Width at 90% of peak (ns)	208 ± 167	192 ± 141	400 ± 154
Total shaping time (μ s)	1.42 ± 0.09	1.26 ± 0.07	1.96 ± 0.09

Peak-and-Hold Circuit



Peak-and-Hold circuit based on a precision rectifier. The outer feedback path of OA_1 is shown at the top through R , while the local path closes through D_1 .

Transient Response of the Pulse Discrimination Chain



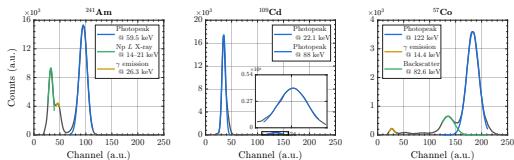
Transient response of the shaper (grey), comparator (green) and P&H (blue) stages to a single ^{241}Am event. The yellow region marks the interval during which the reset transistor is driven HIGH, allowing the hold capacitor to discharge.

Dead Time of the Acquisition Chain

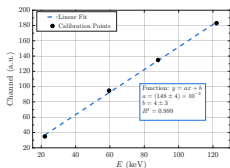
Sampling Condition	Dead Time (μs)	Sampling Protocol
N/A	13.2 ± 0.9	N/A
Internal MCU ADC	59.2 ± 1.5	Analog Read
External ADC	17.8 ± 0.4	SPI

Values are the mean of multiple ^{241}Am events, with the corresponding standard deviation. With a dead time of $17.8 \pm 0.4 \mu\text{s}$, the max. count rate of DORI is $(56 \pm 1) \times 10^3$ counts/s.

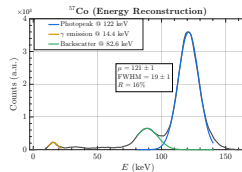
Validation of the Front-End Electronics



(a)



(b)



(c)

(a) Pulse height spectra (CsI(Tl)) for the three calibration sources. (b) Channel-to-energy calibration curve with linear fit. (c) ^{57}Co spectrum reconstructed in energy units.

CsI(Tl) Channel-to-Energy Calibration

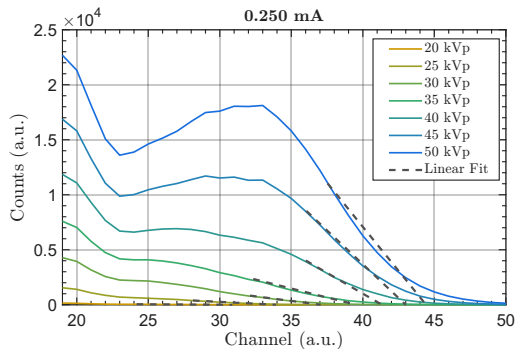
$$C = aE + b$$

- ▶ C – channel number, E – energy in keV
- ▶ Fit yields $R^2 = 0.999$: linear channel-to-energy relationship confirmed up to channel 183 (≈ 590 mV)
- ▶ Validation: ^{57}Co 122 keV photopeak reconstructed at $\mu = 121 \pm 1$ keV, FWHM 19 ± 1 keV $\Rightarrow$ resolution $16 \pm 1\%$
- ▶ Confirms the front-end electronics do not introduce amplitude-dependent errors

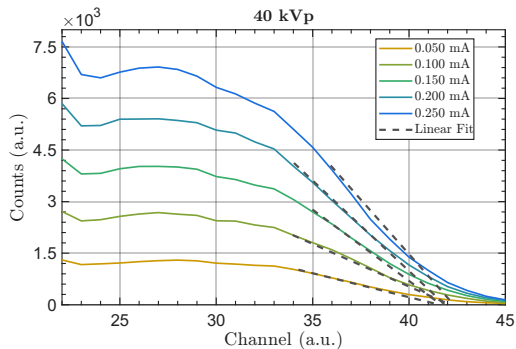
Chapter 3

Calibration of the DORI Prototype

Bremsstrahlung Spectra



(a)



(b)

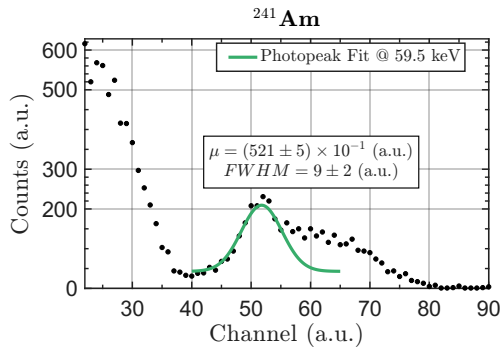
Background subtracted; **(a)** acquired at 0.250 mA for tube voltages 15–50 kVp; **(b)** acquired at 40 kVp for tube currents 0.050–0.250 mA.

Bremsstrahlung Endpoint Channel per Tube Voltage

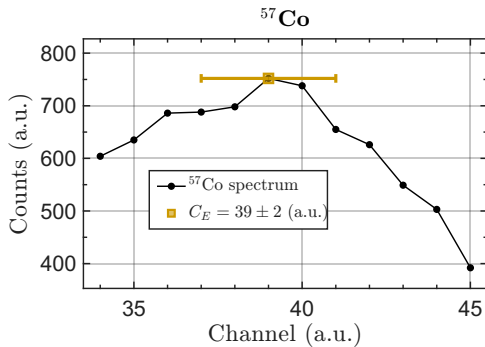
Tube Voltage (kVp)	20	25	30	35	40	45
Endpoint Ch. (a.u.)	31 ± 3%	33 ± 3%	37 ± 1%	39 ± 1%	42 ± 1%	43 ± 1%
		50				
		45 ± 2%				

Values are the mean of five acquisitions with the maximum deviation from the mean.

Reference Source Spectra



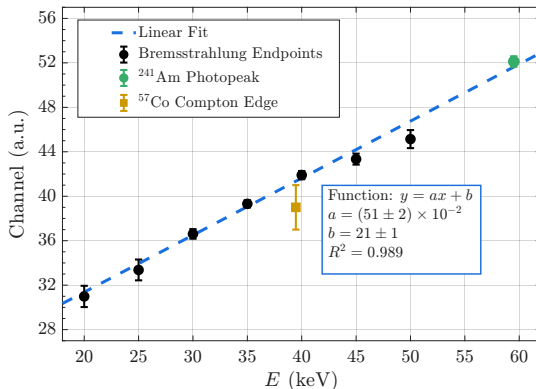
(a)



(b)

Background subtracted, plastic scintillator. (a) ²⁴¹Am, Gaussian fit to the 59.5 keV photopeak. (b) ⁵⁷Co Compton Edge at 39.46 keV.

Energy Calibration of the DORI Prototype



Weighted least-squares energy calibration, obtained from the seven *Bremsstrahlung* endpoints and the ^{241}Am photopeak centroid. The ^{57}Co Compton edge is superimposed for validation.

DORI Energy Calibration Formula

$$C = aE + b, \quad R^2 = 0.989 \quad (20\text{--}60 \text{ keV})$$

$$E = \frac{C - 21}{0.51}$$

$$\sigma_E = \frac{1}{0.51} \sqrt{(0.02)^2 E^2 + 1^2}$$

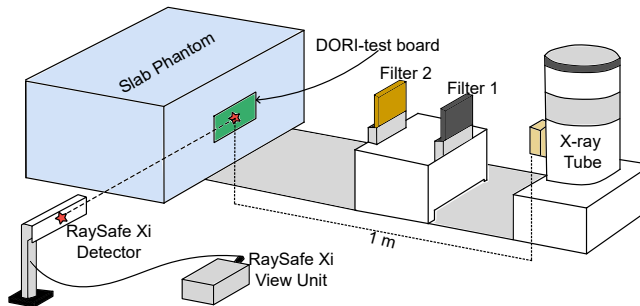
- ▶ ^{241}Am centroid $\Rightarrow$ energy resolution $R = (29 \pm 5)\%$ at 59.5 keV
- ▶ Calibrated interval: 20 to 60 keV (locally linear response)

N-Series Radiation Qualities

Radiation Quality	Tube Potential (kVp)	Added Filtration	$h_{pK}(10)$ (Sv/Gy)
N-25	25	2 mm Al	0.574
N-30	30	4 mm Al	0.815
N-40	40	4 mm Al + 0.21 mm Cu	1.210

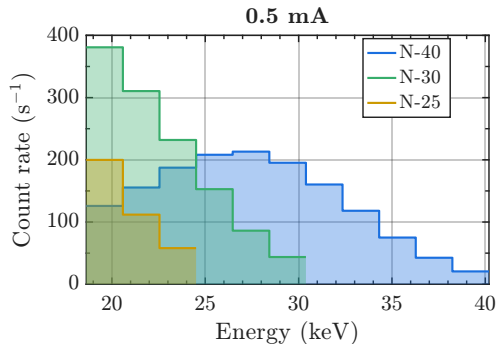
Coefficients correspond to the ISO slab phantom at 1-meter normal incidence (ISO 4037 series).

Dose Calibration Setup

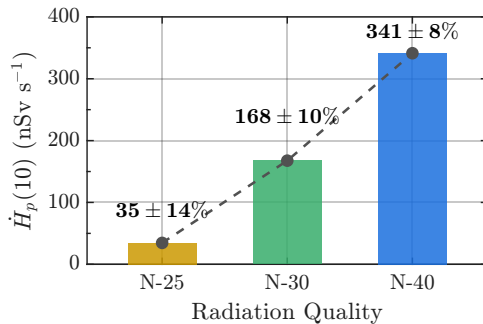


The DORI-test board is placed at the reference measurement point (red star), against the front wall of the built slab phantom. The RaySafe Xi detector is positioned at the same point free-in-air, phantom removed.

ISO 4037 N-Series Spectra & Reference Rate



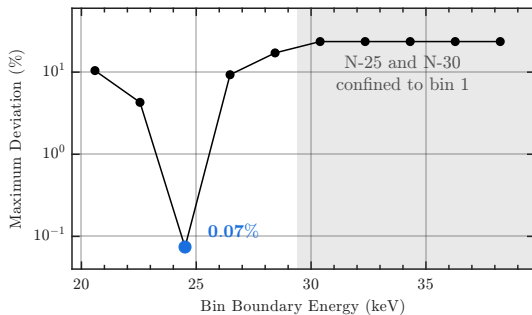
(a)



(b)

N-25, N-30 and N-40, acquired at 0.500 mA for 2 minutes. (a) Energy spectra. (b) Reference $\dot{H}_p(10)$.

Bin-Boundary Optimisation



$$\delta_{\max} = \max_q \left| \frac{H_p(10)_q^{\text{calc}}}{H_p(10)_q^{\text{ref}}} - 1 \right| \times 100\%$$

Minimum $\delta_{\max} = 0.07\%$ at a boundary of 24.5 keV. Bins adopted: $N_1 \in [20, 25]$ keV, $N_2 \in]25, 40]$ keV.

Count Rates per Bin & Reference $\dot{H}_p(10)$

Radiation Quality	N_1 (s ⁻¹)	N_2 (s ⁻¹)	$\dot{H}_p(10)$ (nSv s ⁻¹)
N-25	370	0	35 ± 5
N-30	923	283	168 ± 16
N-40	469	1033	341 ± 26

Dose Calibration Weights & Formula

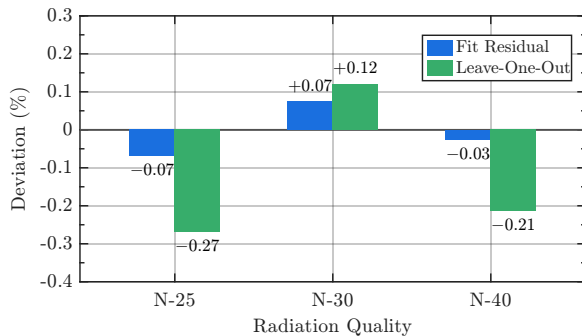
$$w_1 = 93.4 \pm 18.4, \quad w_2 = 287.9 \pm 29.3 \left[\frac{\text{pSv}}{\text{count}} \right]$$

$$H_p(10) = 93.4 N_1 + 287.9 N_2$$

$$\sigma_{H_p(10)} = \sqrt{(18.4 N_1)^2 + (29.3 N_2)^2}$$

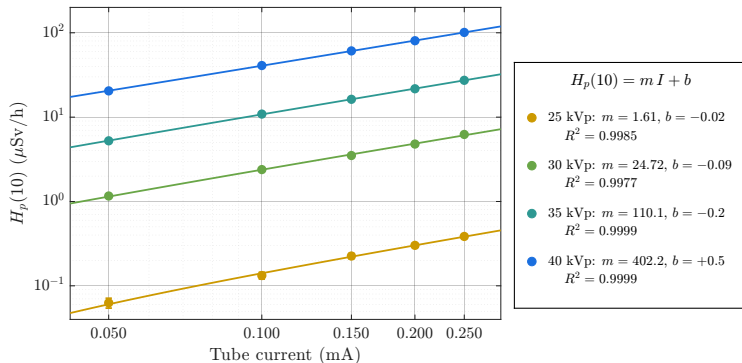
A count in the upper bin carries $3.1\times$ the dose of a count in the lower bin.

Fit Residuals & Leave-One-Out Deviations



Deviation of the reconstructed $\dot{H}_p(10)$ from the reference value, per radiation quality. Blue bars: fit residuals. Green bars: leave-one-out predictions.

Reconstructed $\dot{H}_p(10)$ vs Tube Current

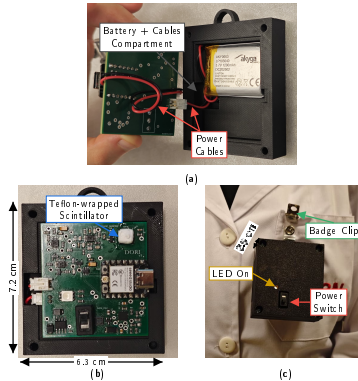


Tube potentials 25–40 kVp, currents 0.050–0.250 mA. Dose rate increases linearly with tube current at every potential ($R^2 \geq 0.998$).

Chapter 4

Final Prototype

Final DORI Prototype



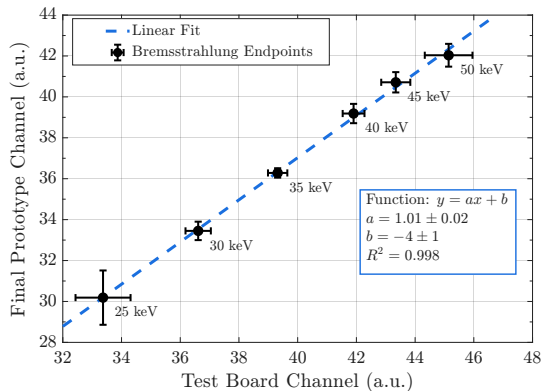
(a) Bottom layer of the enclosure with the LiPo battery compartment. **(b)** Assembled PCB with the Teflon-wrapped scintillator coupled to the SiPM. **(c)** Closed prototype worn over a lab coat, with badge clip, power switch and LED.

Endpoint Channel — Final Prototype

Tube Voltage (kVp)	25	30	35	40	45	50
Endpoint Ch. (a.u.)	$30 \pm 4\%$	$34 \pm 1\%$	$36 \pm 1\%$	$39 \pm 1\%$	$41 \pm 1\%$	$42 \pm 2\%$

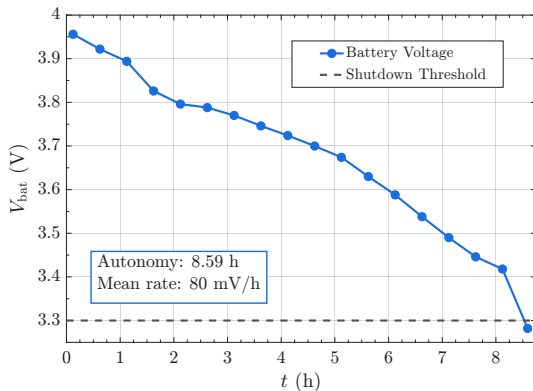
Values are the mean of three acquisitions with the maximum deviation from the mean.

Endpoint Correspondence: Test Board vs Final Prototype



Linear fit, $R^2 = 0.998$, slope 1.01 ± 0.02 : gain preserved. Final prototype records events 3.0 ± 0.2 channels below the test board (baseline shift ≈ 9.6 mV); the energy calibration slope carries over, with only the intercept displaced.

DORI Discharge Curve



Continuous operation at 100 interrupts/second. DORI operated for approximately 8.6 hours before the 3.3 V cutoff, at a mean discharge rate of 80 mV/h.

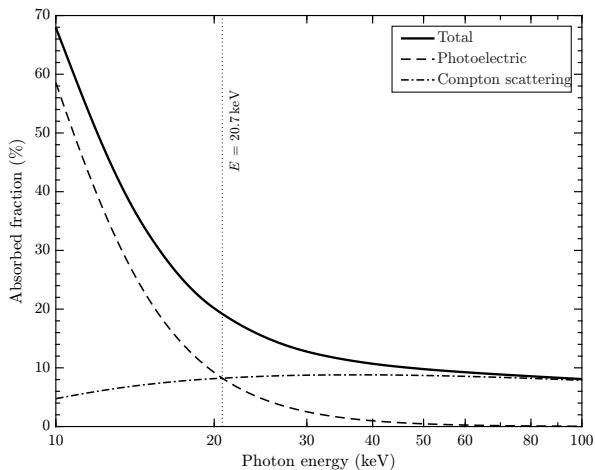
Appendix

Sealed Source Activities

Source	Reference Activity	Reference Date	Serial Number
^{241}Am	TODO	TODO	TODO
^{109}Cd	TODO	TODO	TODO
^{57}Co	TODO	TODO	TODO

Reference activity of the sealed sources used throughout the hardware and calibration chapters. Values to be completed in the thesis.

BC-408 Photon Absorption in 5 mm



Thank you

Marta Alexandra Moniz Rodrigues